

Machine Learning for Natural Language Understanding

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Sources: Lecture and Exercise

Key Content

- Llama 3 (Section 4.1 and 4.2)
 - <https://ai.meta.com/research/publications/the-llama-3-herd-of-models/>
- DeepSeek-V3 (Section 5)
 - <https://arxiv.org/pdf/2412.19437>
- Instruct-GPT
 - <https://arxiv.org/pdf/2203.02155.pdf>

Further Reading

- NLU
 - https://youtu.be/tZ_Jrc_nRJY
(up to minute 51)

Traditional Tasks of NLU

- Relation Extraction
 - https://youtu.be/pO3Jsr31s_Q?t=445
- Grounding
 - https://youtu.be/7b2_3dDTKMc?t=670
- Semantic Parsing
 - <https://youtu.be/C5bdf1sg7rs>
- Question Answering
 - https://youtu.be/NcqfHa0_YmU
- Sentiment Analysis
 - <https://youtu.be/O1Xh3H1uEYY?t=241>
(up to Minute 41)

Flipped Classroom Questions

- How does Post Training differ from Unsupervised Pretraining?
- Name examples for typical NLU benchmarking tasks.
- What is the difference between Supervised Finetuning Data and Reinforcement Learning?
- Name different ways to obtain Supervised Finetuning data.

“The future of language is about the development of language models...”¹

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